Maximizing Binding Capacity for Protein A Chromatography

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Advances in cell culture expression levels in the last two decades have resulted in monoclonal antibody titers of ≥10 g/L to be purified downstream. A high capacity capture step is crucial to prevent purification from being the bottleneck in the manufacturing process. Despite its high cost and other disadvantages, Protein A chromatography still remains the optimal choice for antibody capture due to the excellent selectivity provided by this step. A dual flow loading strategy was used in conjunction with a new generation high capacity Protein A resin to maximize binding capacity without significantly increasing processing time. Optimum conditions were established using a simple empirical Design of Experiment (DOE) based model and verified with a wide panel of antibodies. Dynamic binding capacities of >65 g/L could be achieved under these new conditions, significantly higher by more than one and half times the values that have been typically achieved with Protein A in the past. Furthermore, comparable process performance and product quality was demonstrated for the Protein A step at the increased loading.
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Introduction

Protein A affinity chromatography is currently the industry gold standard for capture and purification of most monoclonal antibodies (mAbs) produced in mammalian cell culture.[1,2] This highly selective capture step forms the basis for the success of the mAb platform purification approach and provides high volume reduction and robust impurity clearance in one step.[3,4] However, with the significant increase in cell culture titers (~5–10 g/L) in the last couple of decades,[5,6] the inherent drawbacks associated with the lower capacity and high cost of Protein A resins become more apparent. With typical loading capacity of most Protein A resins being <40 mg/mL,[7–9] many large scale processes are facing a bottleneck with purifying high titer feed streams.[10,11] Increasing column size to accommodate higher titers leads to exorbitant increases in economic costs.[12,13] Alternatively, increasing number of cycles leads to increase in total processing time which in turn could affect stability of the product in cell culture harvest upon extended hold. Although several attempts have been made in the past to come up with alternatives to Protein A chromatography, the latter still remains the most successful in terms of selectivity.[14–16]

Very recently, a higher capacity Protein A resin (MabSelect SuRe LX) with reported dynamic binding capacities (DBC) at 10% breakthrough of >60 mg/mL_{resin}[17] has been introduced by GE Healthcare. This resin is a successor to the fairly well-established alkali-stable MabSelect SuRe resin but with a more optimized and increased ligand density. However, this resin does suffer from one potential limitation of the need for higher residence time to achieve the optimum high binding capacity.[18] The additional time is needed to allow the antibody to access all the Protein A ligand inside the pores of the bead. The purpose of this study was to evaluate this new generation Protein A resin and determine if an alternative loading strategy could overcome some of the residence time limitations and further push the limits of its capacity.

A dual flowrate loading strategy has been shown in the past to significantly improve productivity in Protein A chromatography.[19] In principle, a faster flowrate (or a lower residence time) was used in the initial stages of loading when all the binding sites are available. Once all of the readily available sites on the surface of the bead matrix are occupied, the flowrate was reduced (i.e., residence time increased) to enable the antibody to diffuse into all the pores and bind to the less readily accessible sites. The time of switch between the two flow rates was also shown to be an important parameter that helped to achieve high capacity while maintaining acceptable processing times. If the switch time occurs too soon, the advantage of having a shorter loading time is not being fully utilized. Alternatively, if the switch time occurs too late, the potential for product breakthrough during loading increases. This concept was validated using a lumped mass transport

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Table 1. Run Design Matrix for DOE Experiments

Factor 1 1st Residence Time (min)	Factor 2 2nd Residence Time (min)	Factor 3 Time of Switch (% of loading time)
5	10.5	75
4	15	75
5	6	50
6	10.5	50
5	15	50
5	10.5	50
6	15	75
5	10.5	25
5	10.5	50
4	15	25
6	15	25
6	6	25
5	10.5	50
4	10.5	50
6	6	75
5	10.5	50
5	10.5	50
4	6	75
5	10.5	50
4	6	25

based chromatographic model and the optimal parameters were calculated using a complex Sequential Quadratic Programming algorithm.[20]

Applying the above mentioned methodology for industrial process development might be impractical and cumbersome as it requires estimation of equilibrium and mass transport parameters as well as access to sophisticated computational capabilities. To circumvent some of these limitations, we introduced the use of an empirical model using a more practical Design of Experiments (DOE) approach for calculating the optimal dual flow rate parameters. Currently, DOE experimentation is gaining in popularity in the industry for process optimization, process characterization and design space generation.[21,22]

In this work, dynamic binding capacities (DBCs) were first calculated on the MabSelect SuRe LX resin and compared to those on MabSelect SuRe over a wide residence time range of 2–12 min. Secondly, using a DOE approach, a response surface model was generated for the dual flow rate loading strategy correlating DBC to the two residence times and time of switch. Then, the optimization tool in the DOE software was utilized to maximize capacity while keeping overall residence time acceptable. The predicted condition was subsequently verified across a wide range of antibodies. Lastly, preparative experiments were performed to examine the impact of higher loading on impurity clearance on MabSelect SuRe LX using identical process conditions as MabSelect SuRe.

Materials and Methods

Materials

Monoclonal antibodies (mAbs) A, B, C, D, and E were used as model proteins for this study. They are all recombinant humanized monoclonal IgG_1 antibodies produced in Chinese hamster ovary (CHO) cells at Biogen Idec.

Load material for breakthrough studies was purified drug substance diluted with equilibration buffer (75 mM sodium phosphate, 100 mM sodium chloride, pH 7.3) to 5 ± 1 mg/mL to mimic typical harvested cell culture fluid (CCF) conditions. CCF was used as the load material for preparative runs.

All buffer chemical components used were purchased from Sigma (St. Louis, MO) and J. T. Baker (Phillipsburg, NJ) unless stated otherwise.

An ÄKTA Explorer 100 purification station from GE Healthcare was used for all chromatographic experiments.

For the purpose of conserving material, columns used for breakthrough and preparative experiments were packed into 6.6-mm diameter Omnifit glass columns (Diba Industries, Danbury, CT) to a bed height of 10 ± 2 cm.

Experimental Methods

Estimation of dynamic binding capacity

Dynamic binding capacity (DBC) was determined from breakthrough curves using frontal experiments. To ensure minimal variability in-between runs, all load material and buffer solutions were prepared in one batch. Purified drug substance diluted to the representative load conditions of a typical Protein A step was applied to a pre-equilibrated column. Loading was stopped when the concentration of mAb in column outlet (monitored using ÄKTA UV_{280} nm) reached 10% of feed concentration. The DBC at 10% breakthrough was then estimated using the following equation:

$$DBC_{10\%} = \frac{(V_F - V_O)C_F}{V_C} \tag{1}$$

where V_F is the volume of protein loaded at 10% breakthrough, V_0 is the hold up volume of the system, C_F is the protein concentration of the feed, and V_C is the column volume.

Once target breakthrough was achieved, the column underwent two wash steps, the bound protein was then eluted from the column followed by column regeneration. DBC experiments were often performed in duplicate and average values were taken. An error margin of ±5% can be expected for these measurements.

DOE experiments

A response DOE model using a central composite design (CCD) was utilized for optimization of the dual flow loading on MabSelect SuRe LX. Three parameters examined in this DOE were 1st residence time (A), 2nd residence time (B), and time of switch (C). Residence time was used in place of linear flow rate to make the model independent of column bed height. The test ranges for A and B was 4–6 and 6–15 min, respectively. C was expressed in terms of % of load time and ranged from 25 to 75%. To make C an independent factor, as required by the design, the load time was based on a base-case scenario of 55 g/L loading at 6 min RT with an antibody titer of 5 g/L. Table 1 shows the design matrix with 20 randomized runs including 6 center points. Six center points were deliberately included to account for the variability in experimental DBC determination.

For each condition mentioned in Table 1, breakthrough curves were generated following the experimental procedure mentioned in the previous section. The main response monitored for each condition was DBC at 10% breakthrough. The time-averaged residence time, referred to as effective residence time (E_{RT}), for each condition was also calculated using the following equation:

$$E_{RT}=(A \times C/100) + ((B \times (1-C)/100)) \quad (2)$$

ANOVA analysis of the experimental results was performed using Design-Expert 8 software (Stat-Ease, Minneapolis, MN). The model was generated with 95% confidence interval.

Preparative Protein A experiments

The impact of higher loading on impurity clearance was assessed by performing side-by-side preparative runs on MabSelect SuRe and MabSelect SuRe LX. MabSelect SuRe was operated under the single flow technique at a 6-min residence time (control). Clarified cell culture fluid of molecules A, B, and E was loaded on to pre-equilibrated columns. Apart from the loading step, all other operating conditions between the two resins were identical. Following load, the columns were washed three times before bound target mAB was eluted from the columns. The columns were then regenerated with a base solution and stored until further use. Eluate pools from each run were adjusted to a neutral pH and submitted for analytical testing to assess the levels of host cell protein (HCP), DNA, and residual leached Protein A (rPA) in the pool.

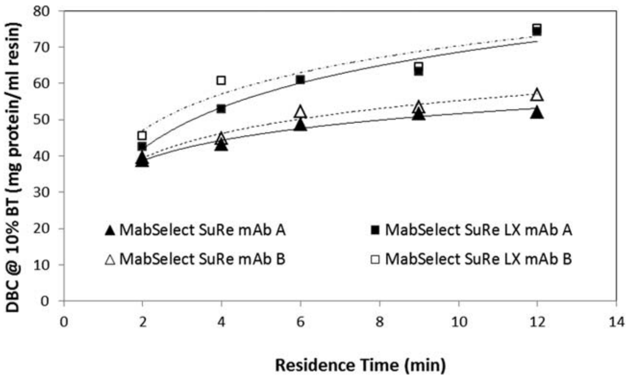

Figure 1. Comparison of DBC of MabSelect SuRe and MabSelect SuRe LX as a function of residence time.

Table 2. ANOVA Analysis of the Statistical Model

Parameter	*P* Value
1st residence time	<0.0001
2nd residence time	<0.0001
Time of switch	<0.0001
R^2	0.9632
Adj. R^2	0.9301
Pred. R^2	0.7757
Adeq. precisior	20.83

Results and Discussion

Comparison of dynamic binding capacity

DBC as a function of residence time was evaluated for MabSelect SuRe and MabSelect SuRe LX using mAbs A and B under identical chromatographic conditions (Figure 1). Comparable DBC was observed for the two resins at lower residence times (2–4 min) but as residence time was increased, the benefit of higher ligand density offered by MabSelect SuRe LX became more apparent. At a 6 min residence time, an ~20% increase in capacity was observed on MabSelect SuRe LX compared to its predecessor MabSelect SuRe for both antibodies. As residence time was increased further from 6 to 12 min, DBC on MabSelect SuRe reached a plateau while an additional ~20% capacity increase could be achieved on the LX resin. These results are in agreement with those reported by the manufacturer.[23] At saturation, the observed maximum DBC was close to 80–90% of the respective maximum equilibrium capacity (data not shown).

Optimization of operating conditions for maximum capacity

The dual flow rate loading strategy was employed on MabSelect SuRe LX to maximize for capacity while trying to minimize overall residence time. The operating parameters

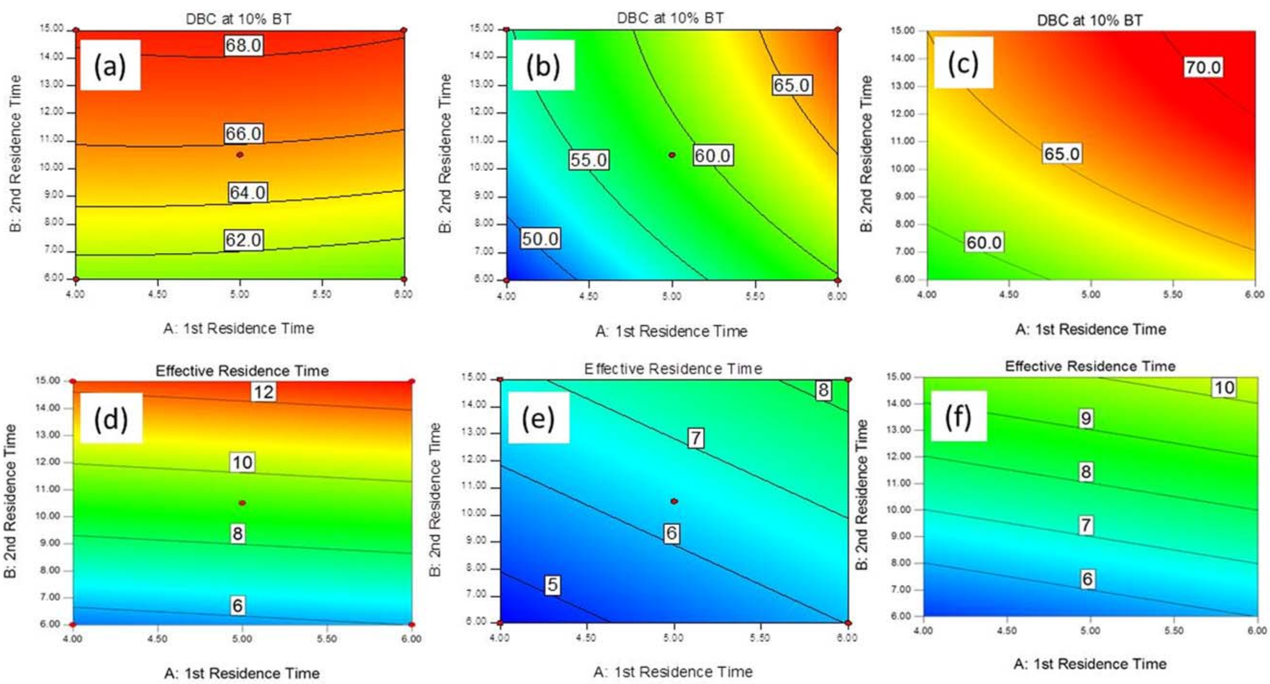

Figure 2. Contour plots showing DBC (a–c) and effective residence time (d–f) as a function of the two residence times at time of switch = 25% (a and d), 75% (b and e), and 50% (c and f).

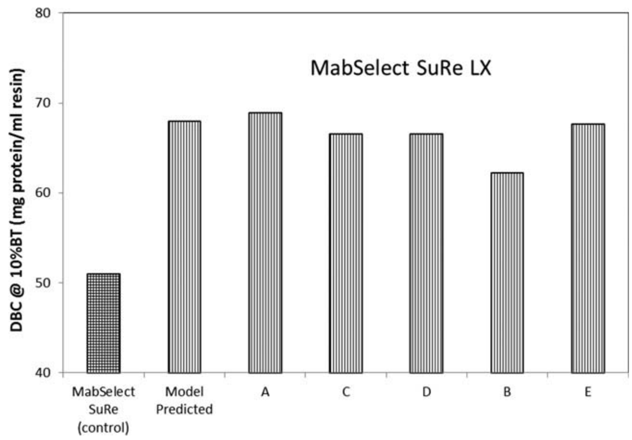

Figure 3. Comparison of DBC using model optimized dual flowrate conditions for mAbs A–E.

DBC on MabSelect SuRe for mAb A is also shown as control.

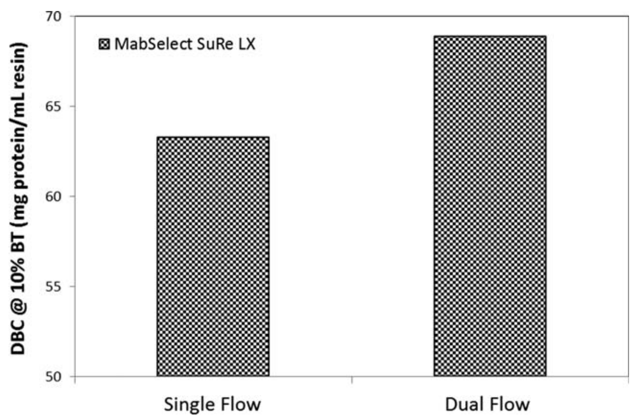

Figure 4. Comparison of DBC using single flowrate vs. dual flowrate loading strategy on MabSelect SuRe LX.

Table 3. Comparison of Preparative Runs on MabSelect SuRe and MabSelect SuRe LX

Molecule	Resin	HCP LRV	DNA LRV	rPA (ppm)
A	MabSelect SuRe	2.0	6.4	3.8
	MabSelect SuRe LX	2.1	5.1	1.2
E	MabSelect SuRe	3.3	6.7	2.5
	MabSelect SuRe LX	2.9	6.7	4.6
B	MabSelect SuRe	2.7	3.0	2.1
	MabSelect SuRe LX	2.6	3.1	3.2

1. All yields >95%. 2. Target loading for MabSelect SuRe column 35 g/L_{resin}. 3. Target loading for MabSelect SuRe LX column 55 g/L_{resin}.

for the dual flow loading strategy were optimized using a DOE approach. As described in "Materials and Methods", a response surface model using a central composite design (CCD) was used. The test ranges for each factor (Table 1) was based on trends observed in Figure 1 and facility considerations. Residence time was selected in place of linear flow rate to make the model independent of column bed height. The lower limit of the first residence time (A) was kept at 4 min due to pump capability limitations in our existing large scale manufacturing facility. Residence times greater than 15 min for the second residence time (B) were not explored since the binding capacity appeared to plateau after 12 min (Figure 1). The time of switch (C) was expressed as percentage of loading time to make it independent of product titer. The DBC at 10% breakthrough was monitored as the main output response and the effective residence time was calculated using Eq. (2).

The data was analyzed using analysis of variance (ANOVA) to evaluate statistical significance of the generated model. Table 2 shows the adjusted R^2 value of the model to be 0.93 and within reasonable agreement to the predicted R^2 value. The adeq precision value comparing the range of predicted values at each design point to the average prediction error indicates the response is unlikely due to noise of the experiments. Model P values for each parameter were <0.0001 indicating all terms to be statistically significant. All three terms showed a 0.01% probability that the difference between results was caused by random noise. Moreover, the model predicted outcomes were in close agreement with the experimentally measured ones.

Contour plots showing 10% DBC and E_{RT} as a function of the three factors are shown in Figure 2a–f. As expected, at early switch times, DBC became fairly independent of A (Figure 2a). The region of maximum capacity was reached at higher values of B but at the expense of longer E_{RT} (compare Figure 2a and 2d). On the other hand, if the switch time was made too late, DBC values were somewhat lower (Figure 2b) since the protein did not get enough time to diffuse into all of the less accessible pores. Optimum DBC was seen at an intermediate value of switch time (Figure 2c) with E_{RT} values lower than what was encountered for the first scenario (compare Figure 2d and 2f).

After generating the response surface model, the optimization algorithm in the Design Expert software was used to maximize binding capacity while maintaining effective residence time at <9 min. As cell culture titers have increased significantly in the last decade resulting in much lower load volumes and thereby loading times, more emphasis has been put on capacity maximization over decreasing load time. Based on model predictions, the optimum conditions were at $A = 6$ min, $B = 14.6$ min, and $C = 65.7\%$. This translated into an effective residence time of 8.9 min. To validate model predictions, DBC for mAb A was experimentally determined under these optimum conditions (Figure 3). As shown in Figure 3, the experimental and model predicted capacities were in excellent agreement to each other. In addition, the same optimum conditions were also used to determine the binding capacity for four other mAbs (B, C, D, and E). Even though the statistical model was generated using mAb A, the data in Figure 3 suggests that the model could be applied reasonably well for other antibodies too (except for mAb B). This was possible likely due to the similarity in transport properties of most monoclonal antibodies. Figure 3 also shows the binding capacity on the control resin (MabSelect SuRe) for comparison.

To further decouple the reason (change in resin vs. change in loading strategy) associated with the increase in capacity observed on MAbSelect SuRe LX, an additional run was performed on this resin using the conventional single flowrate loading strategy at a comparable residence time (9 min). Although the majority of the increase in capacity occurs just from the change in the resin itself, the dual flow strategy increased capacity by an additional 10% (Figure 4). While this difference is relatively small, it is a real trend and slightly more than the error margin of the experimental technique.

Table 4. Comparison of DBC and Productivity on MabSelect SuRe and MabSelect SuRe LX at 6, 9, and 6 min Effective RT

	mAb A		mAb E		mAb B	
	DBC (mg/mL resin)	Productivity (grams of mAb/L resin/day)	DBC (mg/mL resin)	Productivity (grams of mAb/L resin/day)	DBC (mg/mL resin)	Productivity (grams of mAb/L resin/day)
Scenario 1	48	282	47	277	50	291
Scenario 2	63	298	63	299	63	299
Scenario 3	62	344	63	345	61	336

Antibody titers of 5 g/L and 80% of DBC values used for productivity calculations.

Preparative runs for comparison of product quality

The main advantage of using protein A is the high selectivity offered by this mode of chromatography. With over 90% purity achieved from this single step, it serves as the robust capture step for a majority of manufacturing processes. Preparative runs using clarified cell culture fluid were performed on MabSelect SuRe LX to evaluate the impact of higher loading on impurity clearance. Control runs were also performed on MASelect SuRe for comparison. Both resins were loaded at ~70–80% of their DBC at 10% breakthrough. Maintaining this safety factor in loading is a common industry practice to preserve binding capacity over the lifetime of the resin and account for any variations during DBC determination. The optimum dual flowrate loading condition that was defined in previous sections was slightly modified to facilitate ease of transfer into manufacturing settings. Parameters B and C were rounded to more practical values while maintaining similar effective residence time to the initial DOE generated optimum condition. The modified condition (A = 6 min, B = 13 min, C = 60%) generated an effective loading residence time of 8.8 min on MabSelect SuRe LX. Monoclonal antibodies A, B, and E were selected for this study. mAb B was deliberately chosen as it had shown the highest deviation from the model predictions. Other than loading, all other operating conditions were identical between the two resins. Eluate pools were analyzed for yield, HCP, DNA, and residual protein A. The data is summarized in Table 3. As seen from Table 3, step yield and impurity clearance was comparable across both resins. Step yields for all runs were >95% indicating no product loss even at the higher loading. HCP and DNA LRV values varied slightly based on the antibody but can be considered mostly comparable for practical purposes considering the variation of the ELISA based assays. The levels of leached Protein A levels were slighly higher on the new resin (as expected from the higher ligand density[18]) but still low enough to be easily cleared through the subsequent polishing steps. Aggregate levels and stability profiles were comparable between the two resins for all antibodies evaluated (data not shown). It is also to be noted that even though the MabSelect SuRe LX resin was loaded ~50% higher, the elution peak was collected to the same CV thereby resulting in significantly higher elution pool concentration and lower pool volumes for the subsequent downstream process. This is of tremendous advantage in high titer antibody processes where large process pool volumes often become the bottleneck in the manufacturing facility.[11]

Comparison of binding capacity under similar residence times

One of the primary benefits of utilizing the dual flowrate loading strategy on the MabSelect SuRe LX resin is the ability to realize higher binding capacity without sacrificing cycle time. To illustrate this point further, Table 4 shows the dynamic binding capacity and productivity for three scenarios: (i) MabSelect SuRe at 6 min RT using single flow rate, (ii) MabSelect SuRe LX at 9 min RT using dual flowrate, and (iii) MabSelect SuRe LX at 6 min effective RT using dual flowrate (50% of loading was done at 3 min RT and 50% at 9 min RT resulting in a 6 min effective RT). Productivity or throughput was calculated as 'gms mAb/L resin/day' to make it independent of column dimensions. Table 4 clearly shows that the dual flowrate at 6 min is able to achieve similar capacity as 9 min on the LX resin. This means that the loading strategy is able to achieve the binding capacity gains of LX while still maintaining the processing times of MAbSelect SuRe which has an immediate impact on productivity. The increase in productivity by the resin change alone is significantly less than the increase achieved by the synergistic combination of the dual flowrate and the resin change.

Conclusion

With increasingly higher cell culture titers for monoclonal antibodies, there is a pressing need to develop a higher capacity Protein A capture step to prevent a bottleneck in the downstream process. This work herein demonstrates a practical way of achieving very high DBC (>65 g/L) on Protein A to address the demands of high titer processes. Previous work in the literature had introduced the concept of dual flow loading strategy to increase both productivity and capacity of protein A resins.[19] This work extends the possibility of applying this loading strategy on a new generation protein A media (MabSelect SuRe LX) in an effort to maximize binding capacity on Protein A. A simpler DOE based empirical model was used for the optimization of operating conditions in contrast to the complex mass-transport model based optimization algorithm used in the past. The model was built using one mAb but successfully applied to others mAbs as well.

Compared to its predecessor resin MabSelect SuRe, MabSelect SuRe LX can provide a significant increase in capacity but at the expense of longer residence times. The dual flow loading strategy can help to achieve the same high binding capacity on this new resin without any significant sacrifice in processing time which can result in significant improvement in productivity. Under optimized conditions, comparable process performance and product quality was shown at more than one and half times higher loading. This can translate into a 50% reduction in cycle numbers for the capture step and significantly lower in-process pool volumes. Furthermore, it is to be noted that the optimum effective residence time could have been decreased even further if the lower limit of the first residence time was not restricted to 4 min. For facilities that are not limited by flowrate capabilities, this strategy can be further optimized for maximum productivity.

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